

Problem Set 5

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3. a)

The FREQ Procedure

Table of phys1 by phys2

phys1(Physician 1)		phys2(Physician 2)				Total
Frequency	Mild	Moderate	Severe	Very Sev	ere	
Mild	35	10	1	0		46
Moderate	20	32	13	4		69
Severe	6	13	24	6		49
Very Severe	3	7	8	8		26
Total	64	62	46	18		190

Statistics for Table of phys1 by phys2

Kappa Statistics

Statistic	Estimate	Standard Error	95% Confidence Limits	
Simple Kappa	0.3390	0.0495	0.2420	0.4359
Weighted Kappa	0.4433	0.0476	0.3499	0.5366

Sample Size = 190

$\hat{\kappa}_w = 0.4433$ with confidence limits (0.3499, 0.5366). This test statistics indicates moderate agreement. The confidence bounds do not contain 0 so we reject the null that κ is 0 at the $\alpha = 0.05$ significance level.

b)

Symmetry Test

Chi-Square	DF	Pr > ChiSq
11.0087	6	0.0881

The null hypothesis of this test is that the square table is symmetric, i.e. the cell probabilities p_{ij} and p_{ji} are equal. Since the p-value is > 0.05 , we fail to reject the null and conclude that the cell probabilities are equal.